

ImageNet-R-50 Full-Union LogT Prediction Integrator

Run: fd5cb0502d705bbd9662e197e9d867fda2b6c1b633c340f513be4b33579fd8b4
State: COMPLETE

Abstract

We evaluate a capacity-one LogT hierarchy with full-union parent retraining and a persistent prediction integrator on a fixed ImageNet-R-50 split. The integrator reaches 65.567% final and 73.015% mean stage accuracy. A new post-hoc curve trains one fresh rank-16 joint model using exactly the tasks available at each stage, reaching 81.630% mean stage accuracy. Training one joint model through task 50 and retrospectively restricting its output rows changes the mean by +3.164 points, measuring the combined effect of future-task training under the joint architecture. Same-stage gaps at one-node frontiers remain, so future information is only one contributor; optimization path and shared-representation supervision are independently implicated.

A direct 200-way residual integrator observes frozen LogT node behavior. Capacity-one parents retrain on every represented training example; only persistent integrator history is bounded. Accuracy never gates execution.

Primary benchmark comparison

condition	Last (%)	Mean stage accuracy (%)	role
persistent LogT integrator (H=2048)	65.567	73.015	evaluated method
joint-IID, stage-matched	78.867	81.630	post-hoc available-data control
joint-IID, trained through task 50	78.867	84.795	offline ceiling; earlier points are future-informed
Local E2-LoRA	78.100	82.995	secondary reference
Published E2-LoRA	78.580	83.960	external context

descriptive difference	Last (pp)	Mean-stage (pp)
Persistent LogT – stage-matched joint	-13.300	-8.615
True-node oracle – stage-matched joint	-2.133	-1.529
Full-50 joint – stage-matched joint	+0.000	+3.164
Persistent LogT – full-50 joint	-13.300	-11.780
Integrator – local E2-LoRA	-12.533	-9.980

The task-50 offline joint-IID result is the primary ceiling. Earlier evaluations of that model are future-informed references. Local E2-LoRA is secondary; all differences are descriptive.

Future-information diagnostic

The stage-matched control fits a fresh joint rank-16 LoRA adapter and affine head for five epochs at every stage, using exactly the training examples from tasks seen by that stage. The hierarchy nodes and both joint controls use the same pinned ViT backbone, rank and alpha 16, and attention-QKV plus MLP-fc1 adapter targets. Task 50 reuses and re-evaluates the authenticated offline model, providing an endpoint identity check. Test labels never influence fitting or model choice. The locked hierarchy and both joint curves use the same complete 24,000-image training population; the 19,200/4,800 split is confined to the separate clean-development panel.

Full-50 joint minus stage-matched joint is a task-horizon contrast within one joint-training recipe. It bundles later-task examples, their extra global-softmax competitors, and the additional optimizer updates those examples induce; it is not a pure causal estimate of semantic ‘future information.’ The compact table shows task 1, every power-of-two frontier, and task 50. Power-of-two frontiers contain exactly one hierarchy node, so they remove routing and multi-node score-combination effects. The figure and machine-readable tables retain all 50 stages.

task s	true-node oracle	joint-IID, stage- matched		joint-IID, trained through task 50	future-data effect (pp)	matched joint – oracle (pp)
1	95.763	94.915		97.458	+2.542	-0.847
2	88.710	88.710		97.177	+8.468	+0.000
4	84.035	85.614		93.860	+8.246	+1.579
8	81.801	84.803		91.651	+6.848	+3.002
16	77.789	82.396		86.324	+3.928	+4.607
32	74.917	80.276		81.577	+1.301	+5.359
50	76.733	78.867		78.867	+0.000	+2.133
fresh models		training presentations		optimizer steps	training minutes	peak VRAM (GiB)
49		3020210		47330	77.61	3.86
live hierarchy nodes		stages	mean matched joint – oracle (pp)			mean future-data effect (pp)
1		6	+2.283			+5.222
2		15	+1.514			+4.357
3		18	+1.586			+2.524
4		9	+1.046			+1.674
5		2	+1.027			+0.513

At task 50 the two joint controls are the same authenticated model, so their difference is +0.000 point by construction, while it exceeds the true-node oracle by 2.133 points. This rules out tasks beyond the benchmark horizon, but not node-local missing transfer: the final older interval nodes were frozen before later intervals arrived. Before task 50, training through all 50 tasks changes prefix accuracy by +3.229 points on average (range -0.441 to +9.169); the contrast is positive at 45 stages and negative at 4. At the one-node power-of-two frontiers, where routing and frontier fragmentation disappear, the stage-matched joint-minus-oracle gaps average +2.283 points; those same-stage gaps cannot be caused by unseen later tasks. The task-32 row is also an exact decomposition of the oldest node retained at task 50 (tasks 1–32): later-task co-training changes that interval by +1.301 points, while the matched fresh joint model differs from the hierarchy parent by +5.359 points. The live-node grouping is descriptive rather than causal because node count is correlated with task stage.

Task-32 recipe evidence

task-32 condition	classifier initialization	weight decay	presentations	optimizer steps	test accuracy (%)
capacity-one hierarchy parent	unioned child classifier rows	0	78255	1225	74.917
joint-IID, stage-matched	fresh deterministic prefix head	5e-4	78255	1225	80.276

Both task-32 fits receive the same training rows, presentations, and optimizer-step budget, yet the hierarchy parent tests 5.359 points worse. Fewer examples or steps cannot explain this contrast. Convergence and under-training remain possible because

classifier initialization, regularization, and deterministic seed/order differ; they require the proposed factorial and epoch sweep. The recorded final-loss field is a last-minibatch diagnostic and is deliberately not compared.

Hierarchy-compression context

condition	live adapters/models	Last (%)	role
all-leaf true-task oracle	50	93.117	no consolidation; diagnostic task identity
capacity-two retrained true-node oracle	8	79.350	less-compressed source hierarchy
capacity-one full-union true-node oracle	3	76.733	current hierarchy
joint-IID, trained through task 50	1	78.867	one shared adapter and global head

The no-consolidation leaf oracle reaches 93.117%. The capacity-two retrained hierarchy loses 13.767 points but still exceeds the offline joint model by +0.483 points. Compressing further to the current capacity-one hierarchy loses another 2.617 points and finishes -2.133 points relative to joint-IID. Thus neither true-node routing nor rank-16 LoRA inherently prevents joint-level performance; the evidence points most strongly to accuracy lost as represented intervals are enlarged and consolidated, together with parent optimization. This is descriptive, not a clean capacity ablation: the oracle candidate sets shrink as more nodes are kept; the capacity-two parents used $5e-4$ weight decay and distinct deterministic initialization/order seeds, whereas the current capacity-one parents used zero weight decay.

At task 50 the true-node oracle is given the correct one of three hierarchy nodes and predicts within only 128, 64, or 8 owned classes, whereas the joint model must choose among all 200 classes. The oracle therefore has an easier decision problem. Finishing below joint-IID cannot be blamed on routing and is direct evidence that the retained parent representations or classifier rows are weaker; it is not evidence that label-aware routing itself is harmful.

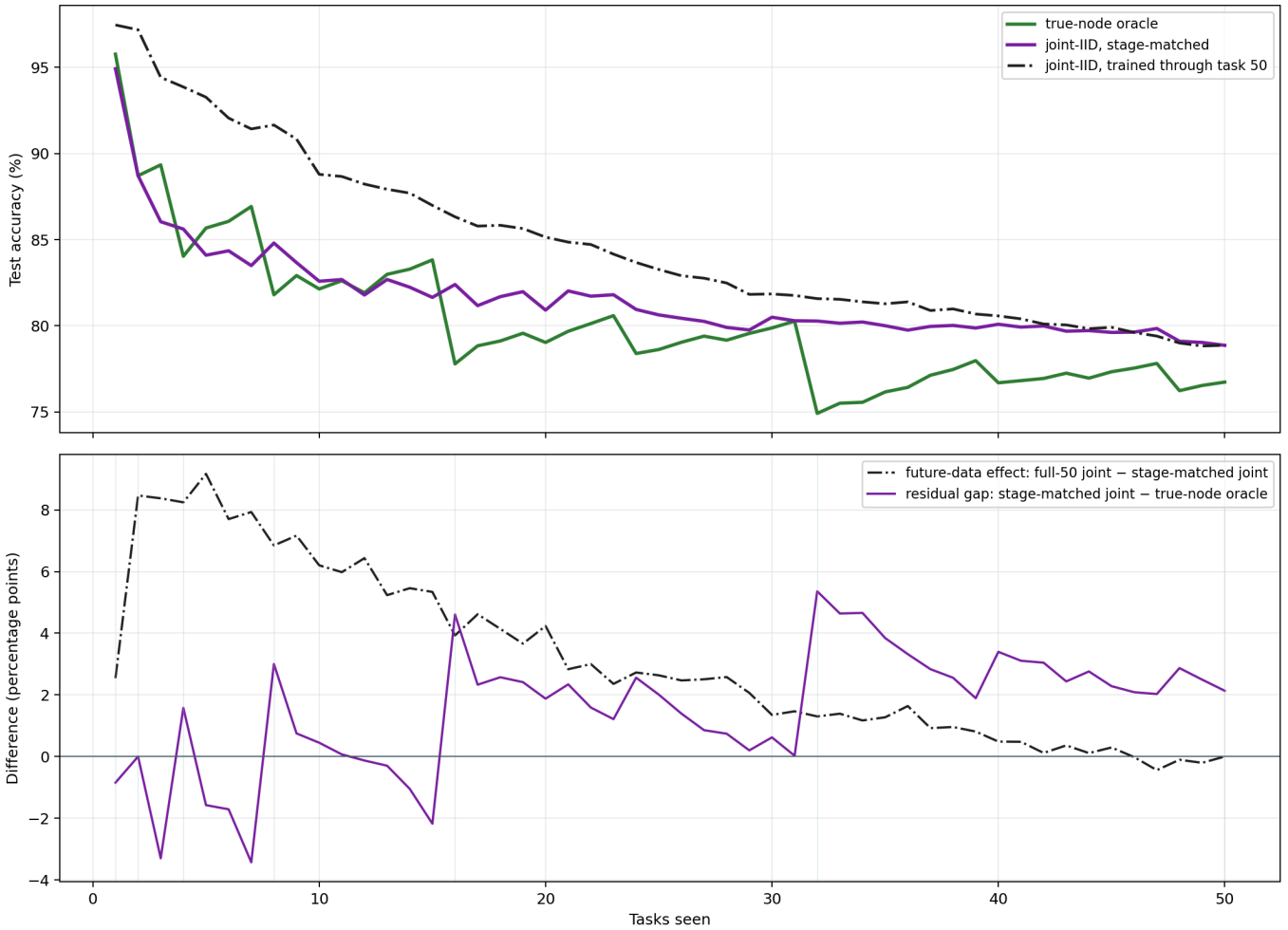
The residual candidate explanations are shared-representation and global-softmax supervision in joint training; union-head initialization and repeated carry optimization in the hierarchy; zero hierarchy weight decay versus $5e-4$ in the joint control; frontier fragmentation and node-local missing later-task transfer at non-power-of-two stages; and single-seed or fixed-budget optimization variance.

Discriminating next experiments

experiment	hypothesis	comparison	decisive value
One-node parent-recipe factorial	union-head initialization, weight decay, order seed, or training budget	At tasks 16 and 32, vary one factor at a time between the parent and stage-matched recipes.	Attributes the routing-free same-data gap instead of treating it as hierarchy loss.
Final-frontier interval decomposition	missing transfer from tasks outside each frozen node	Use the completed task-32 decomposition for tasks 1–32; train fresh offset interval models only for tasks 33–48 and 49–50, then mask full-50 joint to each.	Separates parent-recipe loss from other-task co-training with matched output choices.
Matched capacity-one/capacity-two rebuild	interval size and consolidation severity	Use identical parent initialization, optimizer, seed schedule, and epochs at both capacities.	Tests whether the observed 2.617-point capacity difference survives recipe matching.
Replicated optimization sweep	single-seed variance or five-epoch under-training	Repeat decisive controls over fixed seeds and extend epochs with validation-only stopping.	Adds uncertainty and determines whether gaps are stable or optimization noise.

The adapter-dependent R3 condition targets the separate task-free routing gap. A routing-only change cannot strengthen the owned node used by the true-node diagnostic, although an R3 response integrator could exceed that diagnostic by extracting complementary evidence from non-owning adapters. Parent quality and task-free routing should be measured as separate axes.

What does future training data explain?



Report-only feature diagnostic

feature family	mean validation accuracy
behavior	85.188
behavior_base	85.201
scores	85.194

Configured feature family: **scores**.

Frozen development choices

```
{"feature_variant": "scores", "historical_capacity": 2048, "parent_training": "full_union"}
```

The feature family and H were frozen from the authenticated v2 task-16 development evidence before test access. All displayed fresh and static differences are diagnostic, not thresholds.

stage	fresh full-replay integrator	raw LogT union	true-node oracle	persistent LogT integrator (H=2048)	persistent H=2048 – fresh (pp)
2	88.265	85.714	85.714	88.265	-0.000
4	83.555	83.186	83.186	84.071	+0.516
8	80.958	79.505	79.505	80.801	-0.157
16	77.994	77.284	77.284	77.040	-0.954
50	68.340	66.812	74.521	63.146	-5.194

“Fresh full-replay integrator” is a newly initialized three-hidden-layer residual prediction MLP fit on all prefix observer examples; it is not a fresh LoRA adapter. “Persistent LogT integrator (H=2048)” is the same MLP family warm-started with at most 2,048 historical examples. Validation identities are excluded from all clean trainable components.

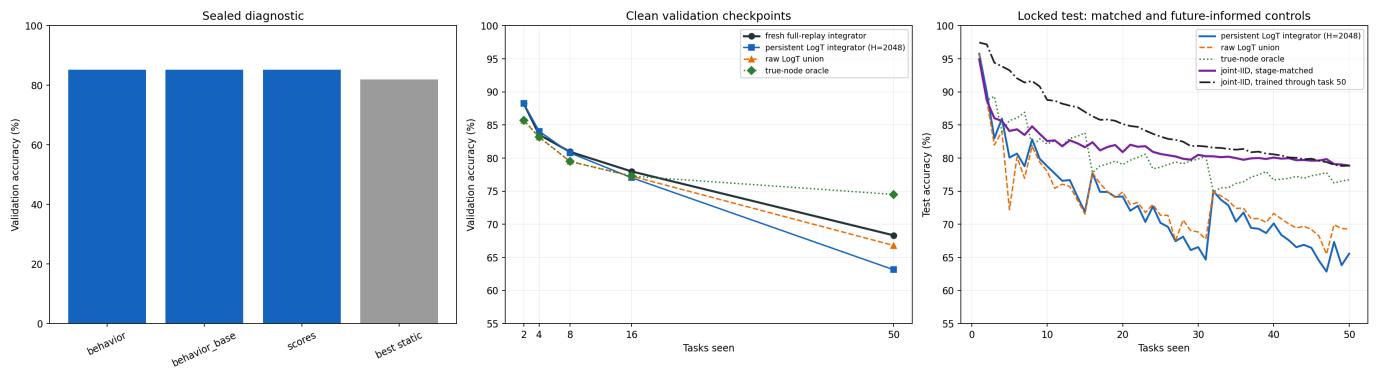
Final-frontier diagnostics

task-free/oracle diagnostic	Last (%)
raw LogT union	69.267
cosine LogT union	67.000
affine-calibrated LogT union	69.567
true-node oracle	76.733

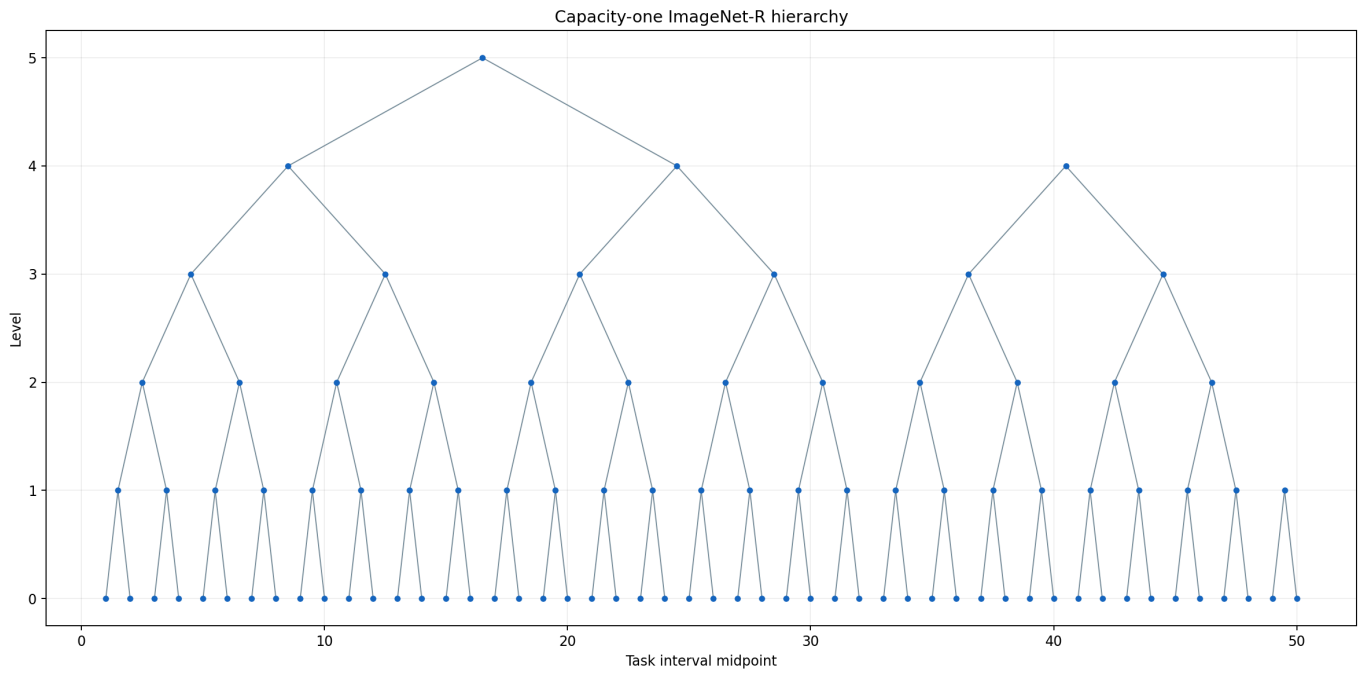
These rows explain the hierarchy frontier and do not define acceptance.

Accuracy

The middle accuracy panel shows fresh full-replay and persistent integrator measurements only at the selected clean-validation checkpoints (tasks 2/4/8/16/50). The right panel shows complete locked-test curves. ‘Joint-IID, stage-matched’ means a separately initialized rank-16 model trained only on data available through that stage. ‘Joint-IID, trained through task 50’ means the one offline model trained on all tasks and then evaluated on each class prefix; its points before task 50 use future training information. Validation and test curves remain in separate panels and must not be compared point-for-point across splits.



Capacity-one lineage



Complexity and resources

There are at most $\text{bit_length}(t)$ carries and $\text{popcount}(t)$ live nodes. Full-union carry cost grows with interval size; cumulative parent presentations are $O(N T \log T)$ for N examples per task. Persistent observer work stays bounded by $\text{popcount}(t) \times (\text{current} + H)$.

condition	node forwards	hierarchy presentations	integrator backward
hierarchy:full_union:fit	—	532280	—
hierarchy:full_union:all_train	—	665325	—
integrator:75d54abc0555..._scores_history2048_fit_seed1993	—	—	462492
integrator:a0eb0c952b80..._scores_history2048_all_train_seed1993	—	—	484988
fresh:clean_fresh:31b06f39e7cf...	—	—	15860
fresh:clean_fresh:4441fcf79df9...	—	—	36540
fresh:clean_fresh:4b656329a6ab...	—	—	131880
fresh:clean_fresh:50dd6999d74c...	—	—	384000
fresh:clean_fresh:729d1e7e749e...	—	—	36540
fresh:clean_fresh:75261357ff55...	—	—	15860
fresh:clean_fresh:7a5de48ba373...	—	—	36540
fresh:clean_fresh:9688456cf3e9...	—	—	15860
fresh:clean_fresh:9b535e6dda45...	—	—	68480
fresh:clean_fresh:b15e63d1bc69...	—	—	131880
fresh:clean_fresh:b20210792e4b...	—	—	384000
fresh:clean_fresh:c1523572ea55...	—	—	68480
fresh:clean_fresh:deca4d0f27c1...	—	—	131880
fresh:clean_fresh:ec682464cdac...	—	—	68480
fresh:clean_fresh:f3b45c65d237...	—	—	384000
fresh:sealed_diagnostic:1923c8b743d9...	—	—	384000
fresh:sealed_diagnostic:354337c25846...	—	—	384000
fresh:sealed_diagnostic:60930840a43c...	—	—	384000
fresh:sealed_diagnostic:66efa248c2ec...	—	—	384000
fresh:sealed_diagnostic:6c1b6aec78fb...	—	—	384000
fresh:sealed_diagnostic:7a8f3effadf0...	—	—	384000
fresh:sealed_diagnostic:7b9344884db9...	—	—	384000
fresh:sealed_diagnostic:b2939a5b1bdb...	—	—	384000
fresh:sealed_diagnostic:b62fa06539b8...	—	—	384000
all_behavior_requests	906760	—	—
shared_behavior_cache	—	—	—